
PH 523: Condensed Matter Physics

Course Code	PH 523
Course Name	Condensed Matter Physics
L-T-P-C	3-0-0-3
Prerequisites	Quantum Mechanics-I and faculty consent.
Intended for	I-Ph.D., M.Sc., B.Tech 3rd and 4th Year
Distribution	Core for I-PhD (Physics) and Elective for others
Approval	9th Senate (May 2016), in the Minor in Physics curriculum

Course Contents

- **Introduction**
- Crystal Structures, Reciprocal Lattice, Brillouin Zones, X-ray diffraction and Structure factor, Defects in Crystal structures [5 Lectures]
- **Lattice Vibrations and Phonons**
- Monoatomic and Diatomic basis, Quantization of elastic waves, Phonon momentum and Phonon density of states, Einstein and Debye model of heat capacity, Thermal properties of solids. [6 Lectures]
- **Electrons in Solids**
- Drude and Sommerfeld theories, Fermi momentum and energy, Fermi surface, Density of states, Electrical conductivity, Ohm's law, Motion in a magnetic field, Hall Effect, Bloch Theorem and crystal momentum, Electron motion in Solids, Kronig-Penney Model, Formation of band, Effective mass [8 Lectures]
- **Semiconductors**
- Intrinsic and extrinsic semiconductors, Acceptor and donor level, Bound State and optical transitions in semiconductors. Degenerate and non-degenerate semiconductor, Optical properties of solids. [6 Lectures]
- **Magnetism**
- Introduction, Origin of magnetism, Bohr-Van Leeuwen theorem, Types of magnetism: Diamagnetism, Paramagnetism, Ferro and Anti-ferro magnetism [7 Lectures]
- **Superconductivity**
- Basic phenomena, Meissner effect, Types of superconductors, London equation, Idea of Cooper pair, Flux quantization, Josephson's tunneling [8]

Text Books

1. C. Kittel, **Introduction to Solid State Physics**, 8th Edition, John Wiley & Sons, 2005.
2. N. W. Ashcroft and N. D. Mermin, **Solid State Physics**.
3. M. P. Marder, **Condensed Matter Physics**, John Wiley & Sons, 2010.

References

1. Phillips, **Advanced Solid State Physics**, Cambridge University Press, 2012.
2. Hook and Hall, **Solid State Physics**, Wiley Science.
3. S. M. Sze, **Physics of Semiconductor Devices**.